Reproduce-then-Extend — Voter Turnout & Term Limits (Political Science) (Day 1)

Intro to Machine Learning · Bruce A. Desmarais

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Published study. Kuhlmann, R. & Lewis, D. C. (2017). "Legislative Term Limits and Voter Turnout." *State Politics & Policy Quarterly* 17(4):372–392. Data: UNC Dataverse doi: 10.15139/S3/POGNLC. The original analysis is an **OLS** regression of state lower-chamber turnout on institutional predictors (Table 2, Col. 1). This example illustrates **out-of-sample prediction with a regression**.

1 Background

Kuhlmann and Lewis ask whether **legislative term limits shape voter turnout** in state legislative elections. Term limits reshape the electoral environment – forcing open-seat contests and changing candidate experience and the information available to voters – with theoretically ambiguous effects on participation. They use **OLS** to estimate the association between the strength of a state's term limits and its lower-chamber turnout, holding constant the major institutional and demographic determinants of turnout (registration rules, professionalism, party competition, election timing, income, diversity). In the reproduced Table 2 model the **term-limitedness coefficient is positive and statistically significant** – more term-limited legislatures see modestly higher turnout – though the largest effects are presidential-election timing and ease of registration.

2 Data and codebook

Unit of analysis: a U.S. state lower-chamber legislative election (a **state-year**); n = 486.

Variable	Definition
`turnout`	lower-chamber election turnout rate, % of the voting-age population **(outcome)**
`tlness`	term-limitedness: strength/bindingness of legislative term limits
`lcr2`	log legislator-to-citizen ratio (per 100,000 residents)
`mmds`	share of legislative seats elected from multimember districts
`regdifficulty`	voter-registration closing date (days before the election)
`straight`	1 if the state offers straight-party-ticket voting
`iuse2`	log number of ballot initiatives on the ballot
`squire`	Squire index of legislative professionalism
`ranney`	Ranney index of interparty electoral competition
`income2`	per-capita income (thousands of dollars)
`diversity3`	Herfindahl index of racial/ethnic diversity
`preselec`	1 in presidential-election years

Variable	Definition
redistrict	1 if districts were newly redrawn (post-redistricting)
year	election year

3 Descriptive results

```
d <- read.delim("data/turnout_state.tab")
mvars <- c("turnout", "tlness", "lcr2", "mmds", "regdifficulty", "straight", "iuse2",
           "squire", "ranney", "income2", "diversity3", "preselec", "redistrict", "year")
d <- as.data.frame(na.omit(d[, mvars]))
preds <- setdiff(mvars, "turnout")

# descriptive statistics
round(t(sapply(d, function(x) c(mean = mean(x), sd = sd(x), min = min(x), max =
↪  max(x)))), 2)
```

```
##                   mean    sd     min     max
## turnout          43.67 10.52   14.67   74.60
## tlness            0.08  0.28    0.00    1.57
## lcr2              1.35  0.90   -1.05    3.61
## mmds              0.16  0.34    0.00    1.00
## regdifficulty    20.81 10.02    0.00   30.00
## straight          0.38  0.49    0.00    1.00
## iuse2             0.77  1.13    0.00    4.00
## squire            0.20  0.13    0.05    0.66
## ranney            0.89  0.09    0.63    1.00
## income2          27.89  8.74   12.53   56.96
## diversity3       33.91 15.55    4.40   73.20
## preselec          0.46  0.50    0.00    1.00
## redistrict        0.22  0.41    0.00    1.00
## year           1998.99  6.87 1988.00 2010.00
```

```
par(mfrow = c(1, 2))
par(mar = c(4, 4, 2.5, 1))
hist(d$turnout, breaks = 25, col = "#a6cee3", border = "white", main = "Outcome:
↪  turnout",
     xlab = "turnout (%)")

# correlation of each predictor with turnout
cors <- sort(sapply(d[preds], function(x) cor(x, d$turnout)))
par(mar = c(4, 8, 2.5, 1))
barplot(cors, horiz = TRUE, las = 1, cex.names = 0.8,
        col = ifelse(cors > 0, "#1f78b4", "#e31a1c"), main = "Correlation with turnout",
        xlab = "correlation")
abline(v = 0, col = "grey50")
```

4 Reproduce the published regression

We fit their exact Table 2, Column 1 specification.

```
ols <- lm(turnout ~ tlness + lcr2 + mmds + regdifficulty + straight + iuse2 + squire +
            ranney + income2 + diversity3 + preselec + redistrict + year, data = d)
```

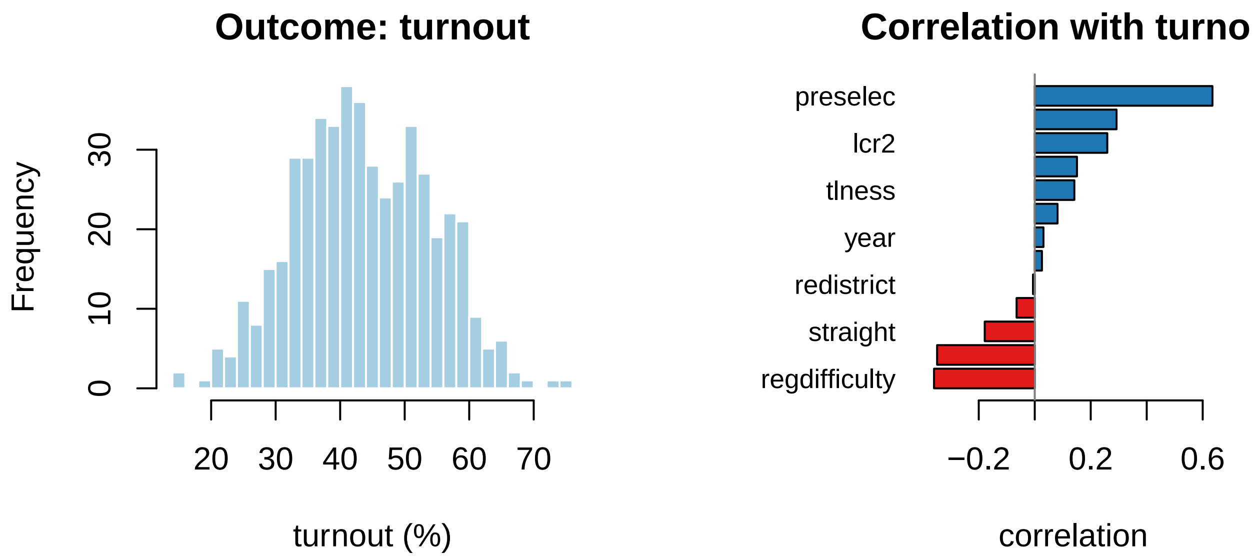

Figure 1: Turnout distribution (left) and each predictor's correlation with turnout (right).

```r
summary(ols)
```

```
##
## Call:
## lm(formula = turnout ~ tlness + lcr2 + mmds + regdifficulty +
##     straight + iuse2 + squire + ranney + income2 + diversity3 +
##     preselec + redistrict + year, data = d)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -20.1833  -3.7146   0.1051   4.0630  16.1266
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)    78.69003  188.82397   0.417  0.67706
## tlness          2.24915    1.09331   2.057  0.04022 *
## lcr2            2.63156    0.44432   5.923 6.11e-09 ***
## mmds           -4.08496    0.82989  -4.922 1.18e-06 ***
## regdifficulty  -0.24137    0.03068  -7.868 2.48e-14 ***
## straight       -1.01880    0.67165  -1.517  0.12997
## iuse2           2.20276    0.27242   8.086 5.24e-15 ***
## squire          9.69863    2.93923   3.300  0.00104 **
## ranney         13.29475    3.14016   4.234 2.76e-05 ***
## income2         0.22176    0.07727   2.870  0.00429 **
## diversity3     -0.13294    0.02256  -5.892 7.26e-09 ***
## preselec       12.78174    0.52126  24.521  < 2e-16 ***
## redistrict      1.44294    0.62707   2.301  0.02182 *
## year           -0.02805    0.09520  -0.295  0.76838
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.612 on 472 degrees of freedom
## Multiple R-squared:  0.723,  Adjusted R-squared:  0.7154
```

```
## F-statistic: 94.79 on 13 and 472 DF,  p-value: < 2.2e-16
```

Confirmation against the published study. Fitting their specification recovers the published model: 486 state-years and in-sample $R^2 = 0.723$. Presidential-election years raise turnout by ~12.8 points (`preselec`), stricter registration deadlines lower it (`regdifficulty`), and professionalism (`squire`) and competition (`ranney`) enter positively — reproducing the reported signs and significance.

5 Out-of-sample prediction with the regression

The same OLS that *explains* turnout can also *predict* it. We hold out a random 30% test set, fit on the other 70%, and evaluate on the unseen states — the core train/test workflow.

```r
set.seed(2026)
train <- sample(nrow(d), floor(0.7 * nrow(d)))
test  <- setdiff(seq_len(nrow(d)), train)

fit    <- lm(turnout ~ ., data = d[train, ])
pred   <- predict(fit, d[test, ])
actual <- d$turnout[test]

# out-of-sample error and R^2 (relative to the training-set mean)
rmse <- sqrt(mean((actual - pred)^2))
r2   <- 1 - sum((actual - pred)^2) / sum((actual - mean(d$turnout[train]))^2)
cat(sprintf("Train n=%d, test n=%d;  out-of-sample RMSE=%.2f points,  out-of-sample
↪  R^2=%.3f\n",
            length(train), length(test), rmse, r2))
```

```
## Train n=340, test n=146;  out-of-sample RMSE=5.83 points,  out-of-sample R^2=0.708
```

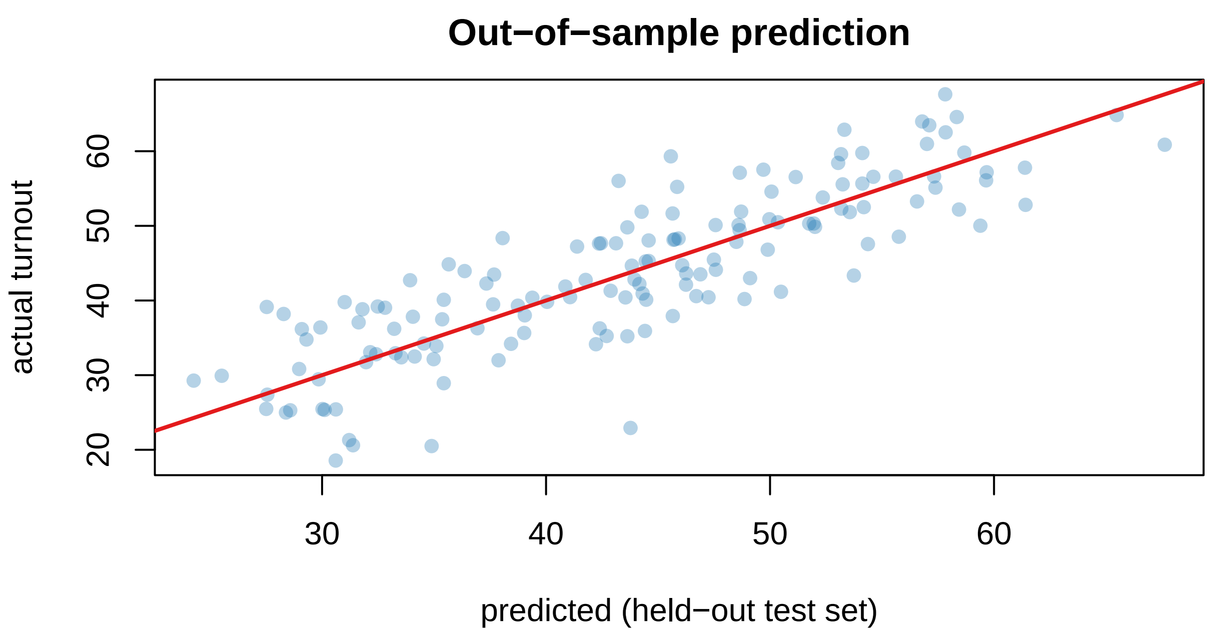

Figure 2: Predicted vs actual turnout on the held-out test set; the red line marks perfect prediction.

The held-out $R^2 \approx 0.71$ is the **honest** estimate of predictive accuracy — scored on data the model never saw, rather than the in-sample $R^2 = 0.72$ that always flatters the fit. The points cluster around the 45° line, so the regression predicts turnout in unseen states reasonably well.

6 Takeaway

The published OLS is the right tool for the paper's *explanatory* question, and the *same model*, scored on a held-out test set, gives a calibrated estimate of how well turnout can be **predicted** out of sample. Explanation and prediction are different goals, and the train/test split is how we measure the second. (Later examples extend this baseline with polynomial terms, regularization, and tree-based learners.)

7 Recommended exercises

1. Re-run the train/test split under several random seeds (or switch to 5-fold cross-validation) and report the spread in out-of-sample RMSE. How stable is the estimate?
2. Add a quadratic term in `income2` (or an interaction between `preselec` and `regdifficulty`) and check whether held-out RMSE improves.
3. Fit a cross-validated lasso on the same predictors and compare its out-of-sample RMSE and its selected variables to OLS.